⚙️ Settings

✅ Using local functions (no API needed)

📊 Dataset Info

- **Users**: 943
- **Videos**: 1,682
- **Ratings**: 100,000

🎬 YouTube Shorts Recommender

Personalized Video Recommendations using Collaborative Filtering

🎯 Recommendations 🔍 Recall Analysis 👤 User Stats ℹ️ About

About the System

🎯 System Overview

This is a collaborative filtering-based recommendation system for YouTube Shorts, using the MovieLens 100K dataset as a proxy.

🛠️ Technology Stack 🔗

- **Backend**: FastAPI + Python
- **Recommendation Algorithm**: SVD (Collaborative Filtering)
- **Recall Strategies**: Multi-path recall (CF, Popularity, High Rating, User Similarity)
- **Frontend**: Streamlit

📊 Dataset Information

- **Users**: 943
- **Videos**: 1,682
- **Ratings**: 100,000
- **Rating Scale**: 1-5

🔌 API Endpoints

- `POST /recommend` - Get personalized recommendations
- `POST /recall` - Get recall analysis
- `GET /user/{user_id}/stats` - User statistics
- `GET /health` - Health check

🚀 Features

- **Multi-path Recall**: Combines multiple recall strategies
- **Personalized Recommendations**: Based on user behavior
- **Cold Start Handling**: Returns popular items for new users
- **Real-time Analysis**: Interactive recall system analysis

📝 License

This project uses the MovieLens 100K dataset, provided by GroupLens Research.